Tech Analyst Assignment Report: Insights from PwC and KPMG Articles

---

Approach - Data Scraping

1. Initial Exploration of Website Structure

Before implementing the data scraping process, a manual review of both PwC and KPMG websites was conducted to understand their structure and how they presented their articles. The observations for each firm are outlined below:

---

2. Observations and Scraping Approach

PwC:

- Key Observations:
  - Dates of articles were clearly mentioned on the main page.
  - A “Load More” button was present, which needed to be clicked programmatically to load all articles with publishing dates in March 2025.
  - All article links redirected to PDFs, requiring direct pdf handling to extract the content.
- Scraping Implementation:
  - Used Selenium to handle dynamic elements like the “Load More” button and to scrape data effectively.
  - Extracted metadata such as URL, title, date, and content (from PDFs).
  - Stored the extracted data in a JSON format for further analysis.

KPMG:

- Key Observations:

    - Articles were categorized into three sections: Insights , News , and Full-Fledged Reports .
    - Dates were not explicitly mentioned on the main page, making it challenging to filter articles by recency.
    - Due to limited availability of March 2025 articles, some articles from April 2025 were included to ensure sufficient data for analysis.
    - Article links redirected to web pages, but some pages included a “Download Today” button that contained more detailed insights in PDF format.
- Scraping Implementation:
    - Used Requests and Beautiful Soup to scrape data from static web pages.
    - Handled cases where additional PDF downloads were required to extract detailed insights.
    - Extracted metadata such as URL, title, date, content (from web pages), and supplementary PDF content.

---

3. Data Collection Summary

- Total Articles Collected:
    - PwC: 9 articles
    - KPMG: 8 articles
- Websites Used for Scraping:
    - PwC: https://www.pwc.in/research-insights.html
    - KPMG:
        - https://kpmg.com/us/en/media/news.html
        - https://kpmg.com/us/en/insights-by-industry.html

---

4. Data Storage

- The scraped data was stored in JSON files with the following keys:
    - `url`: The URL of the article.
    - `title`: The title of the article.
    - `date`: The publication date of the article.

    - `content`: The main content of the article (extracted from web pages or PDFs).
    - `pdf_content`: Supplementary content extracted from PDFs (if applicable).

---

5. Data Processing Pipeline

- Two separate Jupyter Notebook (`.ipynb`) files were created to document the data scraping and cleaning processes for both firms.
- The cleaned JSON files were then passed as input to a comprehensive pipeline for analyzing and comparing the articles.

2. Preprocessing and Analysis

Step 0: Installs and Imports

- Installed necessary libraries and downloaded required NLTK resources for text preprocessing.

Step 1: Preprocessing

- Data Loading : Loaded JSON files containing scraped articles from KPMG and PwC.
- Text Cleaning :
    - Combined `content` and `pdf_content` columns into a single column for KPMG dataset.
    - Performed preprocessing steps:
        - Converted text to lowercase.
        - Removed special characters and stopwords.
        - Tokenized the text using `nltk.word_tokenize`.

Step 2: Classify Articles by Industry

- Used a zero-shot classification model (`facebook/bart-large-mnli`) to classify articles into predefined industries (e.g., Technology, Healthcare, Finance).
- Why this model:

  - Zero-shot models are highly versatile and can classify text into predefined categories without requiring labeled training data.
  - BART (Bidirectional and Auto-Regressive Transformer) is a pre-trained model fine-tuned for natural language understanding tasks.
  - The MNLI (Multi-Genre Natural Language Inference) variant is particularly effective for classification tasks.

Step 3: Extract Keywords

- Initialized the KeyBERT model to extract keywords from the cleaned content.
- Extracted the top 10 keywords for each article using KeyBERT
- Why this model:
  - KeyBERT leverages pre-trained embeddings to extract semantically meaningful keywords.
  - It is lightweight, easy to use, and produces high-quality results without requiring extensive preprocessing.
  - KeyBERT uses cosine similarity to identify keywords/phrases that are contextually relevant to the input text.

Step 4: Cluster Themes

- Generated embeddings for the cleaned content using the SentenceTransformer model (`all-MiniLM-L6-v2`), which converts text into dense vector representations.
- Why this model :
  - SentenceTransformer captures semantic meaning, making it ideal for clustering and similarity-based tasks.
  - It is computationally efficient and works well with large datasets.
  - MiniLM is a smaller version of BERT optimized for performance while maintaining high accuracy.
- Applied KMeans clustering to group articles into 5 thematic clusters based on their embeddings.
- Why KMeans:
  - simple yet effective clustering algorithm for grouping similar items based on their embeddings.
  - It allows us to identify thematic clusters within the articles.
  - The number of clusters (`num_clusters=5`) was chosen to get a basic working model but can be adjusted based on domain knowledge or evaluation metrics like silhouette score.

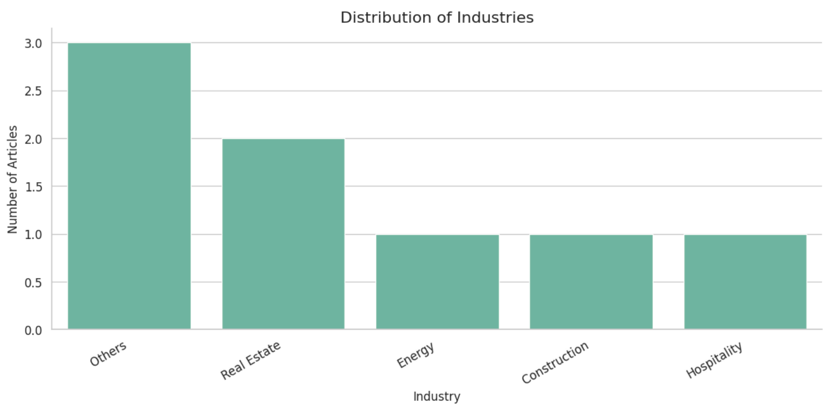
Step 5: Visualizations

- Created visualizations to analyze and compare the results:
  - Industry Distribution: Bar plots showing the distribution of articles across industries for both KPMG and PwC.
  - Cluster Sizes: Bar plots showing the number of articles in each cluster.
  - Comparison Plots: Side-by-side comparisons of industry and cluster distributions between KPMG and PwC.
  - Word Clouds: Visualized the most frequent words and keywords from the articles using word clouds.

---

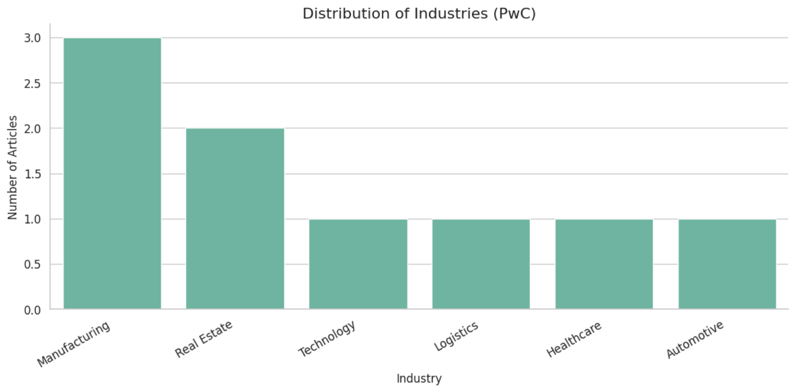
2. Analysis of Vizzes Created

Industry Distribution - KPMG



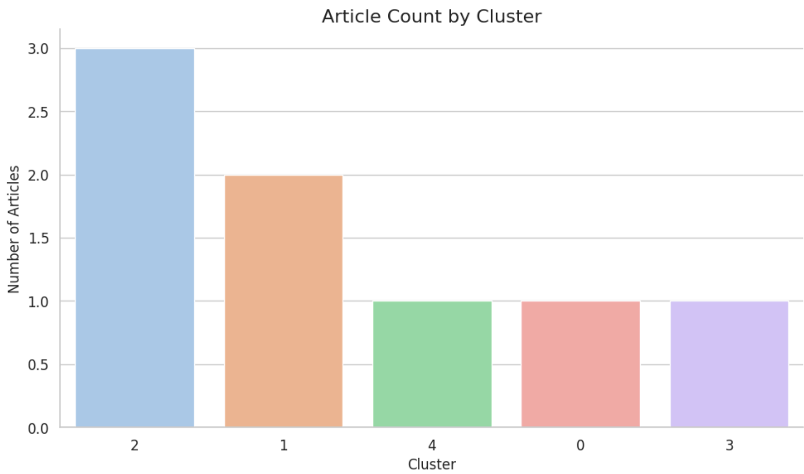
Although, 21 industries were defined earlier, 3 of the articles didn't fit in any. The second highest category is Real Estate.

Industry Distribution - PWC

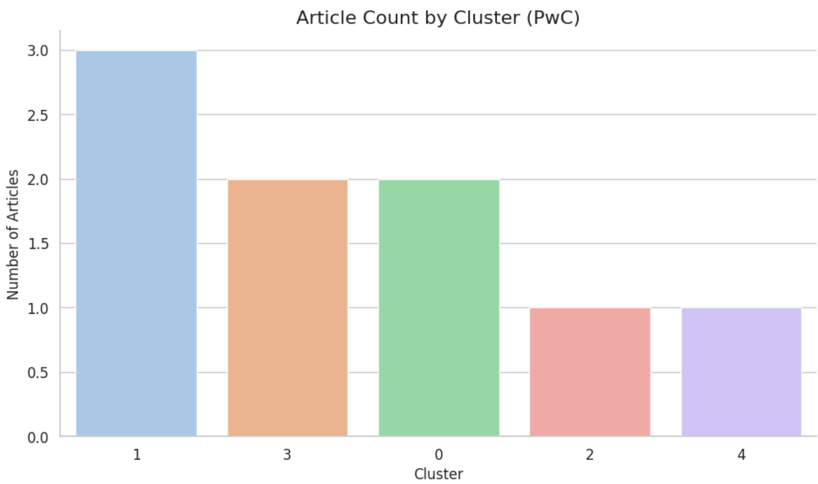


For PwC, Manufacturing is the highest talked about subject in the articles, followed by real estate.

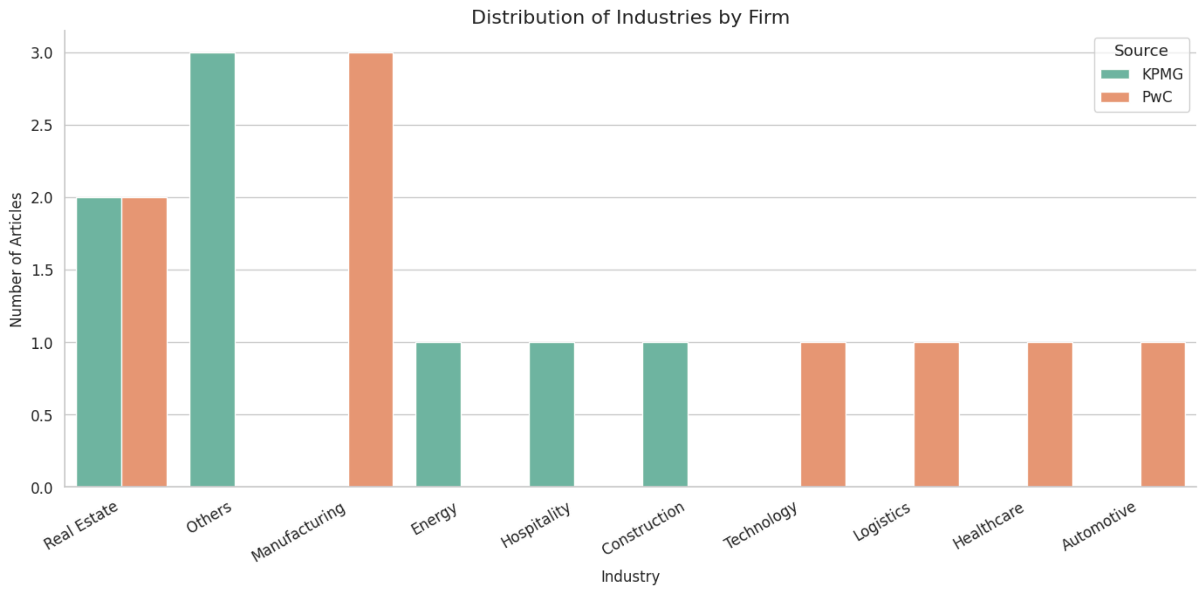
Cluster Sizes - KPMG



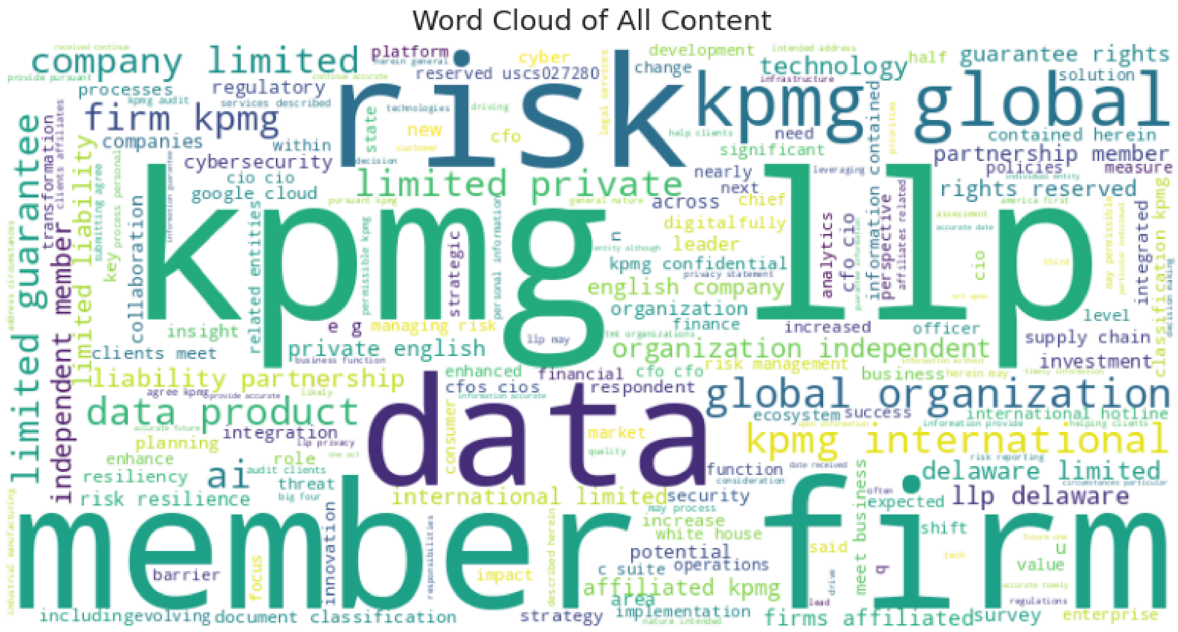
Cluster Sizes - PWC



Comparison Plots



Word Cloud by Whole content - KPMG

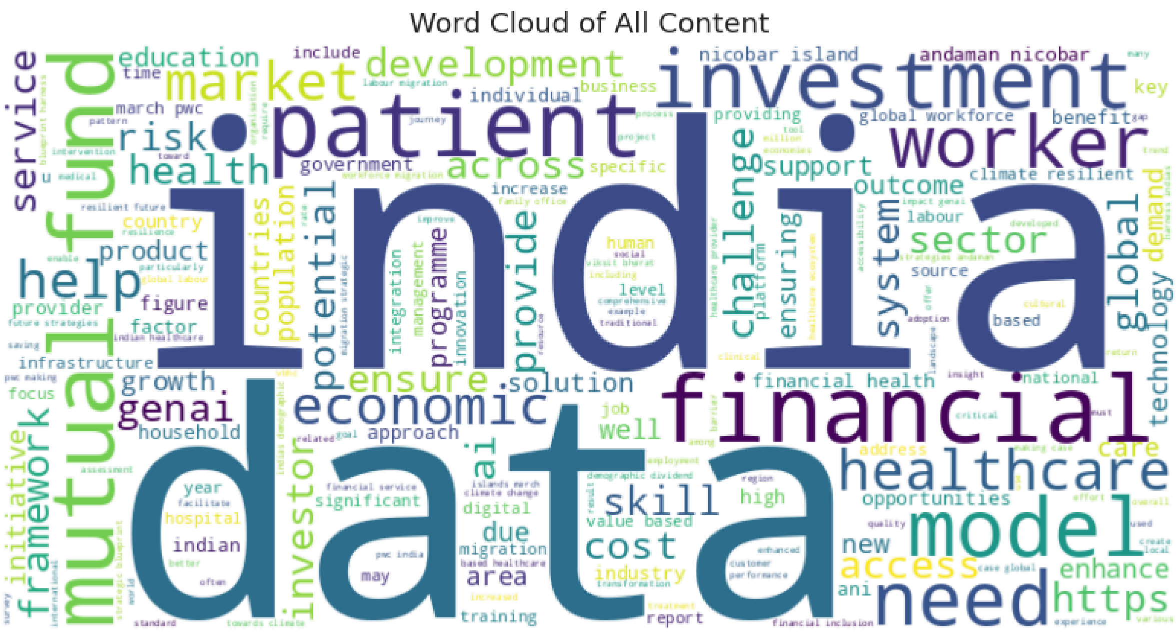


The articles revolve around the firm itself, LLPs, risk, firms and give a very financial perspective to the articles.

Word Cloud by just keywords -KPMG

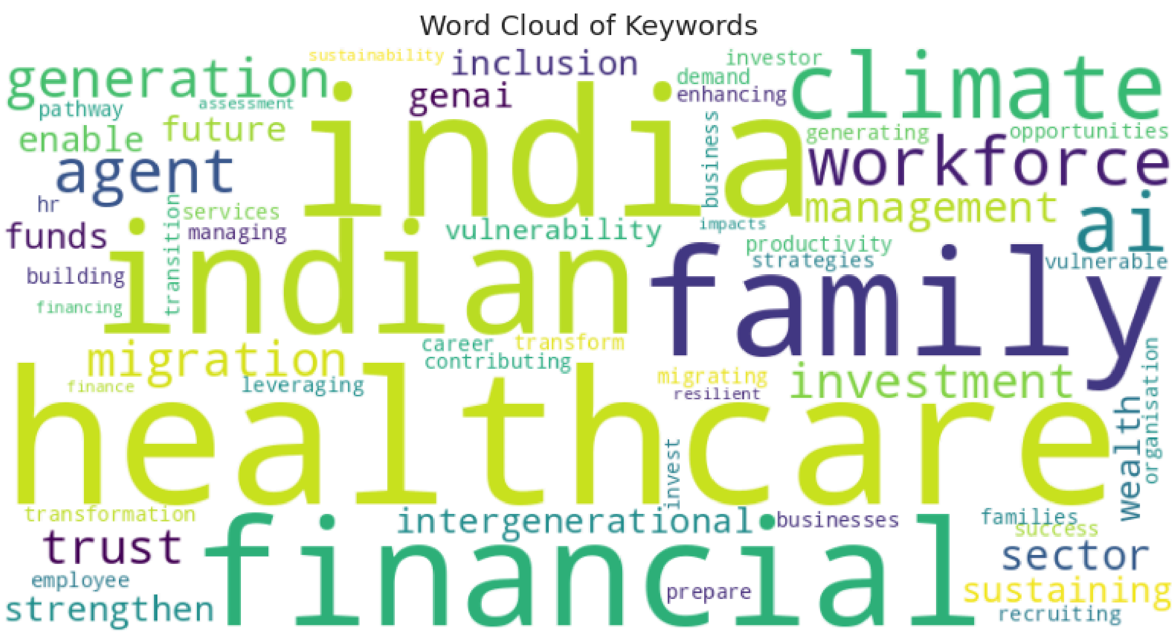
The keywords talked about most are organization, industries, service. (Ignoring Insight)

Word Cloud by Whole content - PWC



The PwC articles, in contrast are very much focused on India and are data centric and deal with a vast variety of topics.

Word Cloud by just keywords - PWC



The extracted keywords highlight the importance of healthcare and family in the country and also touch on financial insights and climate change.

---

3. Further Insights

Industry Focus

- KPMG and PwC both have a strong focus on finance, which reflects their expertise in financial advisory and services.
- Technology is another prominent industry, indicating the growing importance of digital transformation in consulting.
- Differences in industry distribution highlight the unique strengths and priorities of each firm.

Thematic Clusters

- Clustering reveals recurring themes such as healthcare, sustainability, and innovation.
- These themes align with global trends and client needs, emphasizing the firms' adaptability to emerging challenges.

Key Word analysis and themes

- Keywords extracted using KeyBERT provide a concise summary of the main topics discussed in the articles.
- Some of extracted keywords of KPMG :
  - ['industry insights', 'service insights', 'insights workforce', 'companies achieving', 'opportunities services', 'services firms', 'product initiatives', 'insights data', 'respondents industries', 'industry lines']
  - ['capabilities agents', 'deploys ai', 'agent capabilities', 'agents platform', 'deploying ai', 'industries ai', 'services agents', 'entity endeavor', 'cloud develop', 'agent development']
- Some of extracted keywords of PwC:
  - ['management ai', 'management agent', 'career ai', 'agent management', 'productivity ai', 'strengthen hr', 'recruiting ai', 'ai agent', 'agent ai', 'agent employee']
  - ['india migration', 'migration india', 'indias workforce', 'india workforce', 'workforces indian', 'workforce migration', 'indian workforces', 'migrating indian', 'migration indian', 'indian workforce']
  - ['investment family', 'invest family', 'indian families', 'opportunities family', 'wealth family', 'family businesses']

Theme	PwC Insights	KPMG Insights

Financial Inclusion	Highlights the need to expand mutual fund participation across urban and rural populations.	Focuses on leveraging technology to democratize access to financial services globally.
Technology & Innovation	Advocates for simplifying processes through technology and fostering trust via transparency.	Emphasizes AI, data analytics, and blockchain to enhance operational efficiency and client value.
Sustainability	Stresses the importance of aligning investments with sustainable economic growth.	Provides actionable frameworks for integrating ESG principles into business operations.
Regulatory Compliance	Calls for collaboration between SEBI, asset managers, and distributors to ensure investor protection.	Discusses global regulatory changes and their impact on industries like banking, insurance, and energy.

Similarities

1. Focus on Technology :
   Both firms highlight the transformative role of technology in reshaping the financial landscape. PwC emphasizes digital-first solutions for mutual funds, while KPMG explores broader applications such as AI, machine learning, and cybersecurity.
2. Commitment to Sustainability :
   Both organizations recognize the growing importance of sustainability and ESG considerations. PwC ties this to long-term wealth creation, whereas KPMG provides detailed implementation strategies for businesses.
3. Stakeholder Collaboration :
   Both reports stress the need for collective efforts among policymakers, regulators, service providers, and investors to achieve systemic change.

Differences

Aspect	PwC	KPMG
Scope	Narrow focus on India's mutual fund industry and its contribution to national development.	Broad coverage across multiple industries, geographies, and functional areas.
Approach	Forward-looking vision with specific targets	Practical, solution-oriented approach with tools, frameworks, and case studies.
Audience	Primarily aimed at Indian policymakers, asset managers, and retail investors.	Targets a global audience, including corporate leaders, auditors, tax professionals, and regulators.

Conclusion

Both PwC and KPMG provide valuable insights into the evolving financial and healthcare ecosystem, albeit with different scopes and approaches. PwC's vision-centric narrative offers a clear path for India's mutual fund industry, while KPMG's diverse content equips stakeholders with practical tools and strategies.